

Received 27 April 2025, accepted 13 May 2025, date of publication 16 May 2025, date of current version 27 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3570815

RESEARCH ARTICLE

Anomaly Detection for PV Modules Using Multi-Modal Data Fusion in Aerial Inspections

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This work was supported in part by the European Union's Horizon Europe-The EU Framework Programme for Research and Innovation, under Grant 101119744 (TALOS).

ABSTRACT Solar energy is a compelling and growing avenue for transitioning towards reliance on sustainable and environmentally friendly energy sources. However, the performance of individual photovoltaic (PV) panels is vulnerable to defects inflicted by weather exposure. The industry currently addresses these challenges by conducting manual inspections at each PV installation, a process that is highly time-consuming, financially burdensome, difficult to scale, prone to human error, and sometimes unfeasible due to topographical constraints. This study aims to develop a solution that detects these defects in real time resorting to an Autonomous Aerial Vehicle (AAV) equipped with for a thermal and a visual sensor. This paper contributes three major advancements to PV defect detection: a robust, standardized dataset built following IEC TS 62446-3 specifications with annotations from a real PV power plant, a novel multi-spectral approach that leverages early fusion of visual and thermal data captured via an AAV, and benchmark performance metrics for defect detection using state-of-the-art AI models. Our implementation uses Real-Time DEtection TRansformer (RT-DETR) and You Only Look Once (YOLO) models, achieving a mean Average Precision@50 (mAP) of 94% for RT-DETR and a mAP@50 of nearly 95% for YOLOv11. These results demonstrate the effectiveness of our approach in real-world settings, addressing a significant operational challenge in PV plant maintenance while establishing new performance benchmarks for automated defect detection in industrial solar installations.

INDEX TERMS

Deep learning, defect detection, photovoltaic panels, AAVs.

I. INTRODUCTION

In the past decade, society has been pushing towards reducing the environmental impacts of human activity, as the effects of human activity have been proven to be detrimental to our way of living [1]. The 29th Conference Of Parties' (COP29) yearbook of global climate action revealed that in 2023 573 GW of new capacity was added globally, with 73% being derived from solar energy. Furthermore, the investment in the green energy sector has been increasing, with 650 financial institutions aligning with the race to

zero¹ initiative, representing approximately two thirds of all global assets under management [2]. The potential of solar power generation surpasses that of other renewable energy technologies, such as wind and biomass-derived energy, by several orders of magnitude [3]. Moreover, solar energy production technologies have seen their cost decrease as the underlying costs and market prices of solar photovoltaic panels have been dramatically reduced [4].

An important part of solar energy production is the maintenance of the PV panels that comprise solar power

The associate editor coordinating the review of this manuscript and approving it for publication was Wai-Keung Fung².

¹The race to zero's campaign mission is to mobilize leadership and financial support from stakeholders or businesses to halve carbon emissions by 2030.

plants. Aiming to further reduce the cost associated with solar power production, the development of a system that ensures an inexpensive and efficient way to detect defects in PV panels must be considered. Additionally, to prolong the lifespan of a PV panel and guarantee its maximum efficiency, the PV panel must be frequently inspected as it can easily be damaged, specially considering the harsh environment PV panels are placed in. The components in solar panels are subject to unforgiving environments that can damage PV panels in multiple ways. Gupta et al. [5] studied environmental factors such as wind, humidity, temperature, and deposition of dust, which revealed that the dust deposition could influence the system performance and shorten its service life. Mellit et al. [6] compiled the defect types and causes of PV plants, the defects included hot spots, bypass diode defects, and junction box defects, which could result in power generation loss and fire risk. Not addressing certain defects, such as dust deposition, bird droppings, and shading, could translate to permanent damage to the PV cells in the form of hotspots, which increases the repair and inspection costs. The industry resorts to manual inspections of the PV panels, which involves a tedious and time-consuming process, making scaling expensive or impossible considering the lack of qualified personnel for the task, and could be impossible due to the topography in which the PV plant is inserted in. The use of AAVs is superior to manually inspecting each PV panel, as it quickly covers a greater number of PV panels compared to manual inspections. Furthermore, using AAVs is relatively free from topographic mistakes [7], is cheaper to maintain, does not impose any downtime on the PV plant, and provides scalability, making it a more desirable solution as opposed to a manual one. This study explores autonomous aerial vehicle (AAV) solutions to collect both IR and visual images of the PV panels, pairing the data collected with a robust, industry-leading, deep learning model. The AI models studied are YOLO [8] and RT-DETR [9], as they are the most suitable models for real time object detection and classification due to their proven accuracy and inference speed, which coupled with their relatively small size, allows them to be ported to an edge device. YOLO's efficiency comes from its highly optimized convolutional layers and lightweight anchor-based design, whereas RT-DETR focuses on removing the need for anchor boxes, relying instead on bipartite matching for prediction alignment. Pairing AAVs and AI models to detect issues in PV panels has been studied before. However, studies only focus on one spectrum, either the visual or IR, ignoring an entire set of defects, which forces the panel to be manually inspected anyway or be susceptible to defects on the spectrum that is left out. Moreover, establishing a causality relationship between defects in different spectrums also becomes impossible, leaving the maintainers of the solar panels guessing whether the defect was caused by the lack of maintenance (i.e. bird drops may lead to hotspots as discussed in II-A) or if it was imposed by the environment. In summary, this article upholds the following contribution:

- Multimodal dataset collected from a real PV power plant with over 4,000 annotated images of solar panels, captured in both the thermal and visual spectra. This dataset was constructed following the IEC TS 62446-3 standard and includes detailed annotations for various types of defects, as well as images of healthy panels. Furthermore, the dataset is structured into three main components: the visual spectrum dataset, the thermal spectrum dataset, and the early fusion dataset.
- Innovative multispectral detection method detecting defects in PV panels through the simultaneous analysis of images in the visible and infrared spectrum. Unlike previous studies that analyzed each spectrum separately, this research combines information from both spectra. The combined analysis allows for the identification of a broader range of defects and the establishment of causal relationships between them.
- Validation of multispectral defect detection resorting to deep learning models, such as RT-DETR and YOLOv11. These were trained and tested achieving promising results with a mean average precision (mAP) of nearly 95% for YOLOv11 and 94% for RT-DETR. These results and their inference times indicate that the models can be used to detect defects in real time with high accuracy, which is critical for effective maintenance of photovoltaic installations.
- The developed solution have the potential to improve the efficiency of photovoltaic panel maintenance, reducing the costs, time, and effort required for inspections. The system detects a variety of defects, including dirt, bird droppings, hotspots, heated junction boxes, and residual damage, which impact the performance and lifespan of solar panels.

The structure of this article is as follows: Section II first highlights the defects found in PV panels and the available datasets, then reviews existing articles that address this issue, discussing their approaches and the datasets resorted to. Section III details the methods used to create a dataset that would satisfy the requirements imposed. Section V reveals the results obtained. Section VI overviews the main advancements made brought by this research.

II. LITERATURE REVIEW

Available literature usually proposes solutions relying on a single spectrum, either the visual spectrum or the InfraRed (IR) spectrum, which hinders the quality of the inspection. Section II-A details to the kinds of defects that typically affect PV panels, considering both the IR and the visual spectrums. Section II-B sheds light on the available datasets for each spectrum used during training and testing of the Artificial Intelligence (AI) model. Section II-C overviews the methods used in studies that aim to solve the presented issue.

A. UNDERSTANDING DEFECTS IN PV PANELS

There are several types of defects associated to PV panels. These include but are not limited to: snail trails, shading,

soiling, vegetation, hot cell, multiple hot cells, short and open circuit, transfer resistance, heated module junction box, broken or cracked cell front glass, direct current (DC) box component and degradation to the *Ethylene Vinyl-Acetate* (EVA) layer placed between cells and the glass layer [10].

Soiling results from the accumulation of dirt, dust, snow, or particles that are capable of blocking sunlight that are carried by the wind and get deposited on the surface of PV panels, which reduces the amount of sunlight the PV panel can attain, leading to lower energy production. Note that snail trails also have the same effect on PV panels. Both of these defects are relatively easy to detect as a simple visual image can be used to detect them [11]. **Hot cells** (or **Hotspot**) can be a result of a variety of things, such as cracks or broken glass, soiling, shading, and bypass diode fault. Soiling, shading, cracks, and bird drops can lead to hotspots as prolonged exposure to soiling can lead to uneven energy production, creating hotspots. A **bypass diode** is responsible for preventing the cells' inverse polarization, so when a fault of this component occurs, the cell becomes reversely polarized and consumes power generated from its neighbors, increasing its temperature, which can lead to burn marks, browning and even fires [12]. These defects can be detected through IR images, but become impossible to detect in the visual spectrum, which makes IR images a valuable asset in detecting defects in PV panels. The standard International Electrotechnical Commission (IEC) TS 62446-3 [13] also mentions the **heated junction box** and the **transfer resistance** defects that can only be identified via thermal images. These defects typically appear as temperature increases with load, and higher loads can be caused by increased contact resistance due to bad soldering or due to a broken ribbon. In the case of the heated junction box it could also be triggered by low-resistance bypass diodes that carry a significant current. Since the standard IEC TS 62446-3 refers to thermal inspection of PV panels, and given the nature of the defects, these defects can only be identified through the use of IR images.

B. AVAILABILITY OF DATASETS

The datasets are the bedrock of deep learning models. In the realm of artificial intelligence, the efficacy and robustness of deep learning strategies hinge profoundly on the quality and relevance of the underlying datasets. While techniques such as data augmentation and balancing are popular for enhancing dataset quality, ensuring that the dataset comprehensively covers a diverse range of real-world scenarios in a consistent and balanced manner is crucial. The dataset required for this project is not easily obtainable as it requires a AAV equipped with both an IR and an RGB high-resolution camera, as well as proper knowledge of the standard IEC TS 62446-3 [13] used in identifying and classifying PV panels defects. Most of the datasets used in the studies, despite providing crucial details about them have not been made available to the public. However, those that have are not adequate for the

project at hand due to the fact that they only encapsulate a single spectrum or don't resort to AAVs to gather the images, which is the case of the Infrared Solar Modules dataset.² The Deep Solar Eye [14] contains power loss labels as opposed to the necessary bounding boxes to identify and classify defects. Other studies like Zhao's et al. [15] and Cavieres' et al. [16] also provide their datasets, but only cover the infrared spectrum, which makes any cross-validation between spectrums impossible. Furthermore, since different datasets labeled the same defect differently, stitching together two different datasets becomes difficult. Additionally, datasets generated to tackle the problem usually encounter the challenge of class imbalance. Certain defects manifest more frequently than others, resulting in their disproportionate representation within the data. Consequently, during model training, these prevalent defects exert a greater influence, potentially introducing bias into the deep neural network. Lastly, the datasets available follow their own standard and not the IEC TS 62446-3 standard, which compromises the quality of labeling and the authenticity of the results.

C. DEFECT DETECTION AND CLASSIFICATION

Michail et al. [17] compiled the results, methods, and contributions of over 170 studies and discovered that there are numerous ways of tackling the inspection of PV modules problem; the most recurring solution was to employ deep learning techniques. Deep learning strategies resort to Convolutional Neural Networks (CNNs) to identify and classify both the defects and the PV modules in a given image, although some of studies have also employed image pre-processing techniques to aid the CNN. On the other hand, machine learning techniques resort to data retrieved from the plant (i.e., power generation metrics and environmental conditions) to generate models capable of detecting issues within the plant. However, these techniques do not locate the defects or classify them. Lastly, a few studies [18], [19] used solely image pre-processing techniques to locate and classify defects. When analyzing these studies, it is crucial to take into account the resolution of the images, the number and type of defects detected, the amount of data points utilized, the method chosen, and the results obtained. As a result, these details have been documented as they greatly affect the results obtained in these studies.

Some studies geared towards detecting defects in PV modules using images gathered through AAVs or other methods focus on a single spectrum of images, either IR or visual images, this section is split into two different segments where the first segment focuses on the studies regarding visual images and the second segment focuses on studies regarding thermal images. Table 1 presents a compilation of all the studies detailed in this section. In this table, the dataset size refers to the number of images used in the AI model aimed at classifying the defects,

²<https://www.kaggle.com/datasets/marcosgabriel/infrared-solar-modules>, accessed on the 12th of November 2024.

without any of the augmentation proposed by the authors. Furthermore, since each study presented its metrics to assess the performance of its models, the performance column refers to the mean Average Precision (mAP) presented by the authors, if available, otherwise values flagged with classification refer to the classification accuracy.

1) VISUAL SPECTRUM

Gao et al. [20] propose a YOLOv8-based model coupled with a Partial Spatial Attention (PSA) mechanism to improve feature extraction and a Multi-Channel Feature Fusion (MCFF) to optimize detection of smaller defects. To validate this solution the authors used the Panel-2 dataset,³ which contains three types of defects: scratches, broken grids and deface, obtaining an mAP@50 of 87.2% and the Solar panel dataset,⁴ which contains three types of defects: bird droppings, cracks and dust, obtaining an mAP@50 of 72%.

Wang et al. [21] used a combination of You Only Look Once (YOLOv3) neural network, Resnet and a Mask R-CNN. Claiming that since the YOLOv3 struggles to detect small objects, a Mask R-CNN with a Resnet and a feature pyramid network (FPN) as backbones will contribute to the speed and accuracy of the model. This study focused on detecting visual defects, including shading, bird drops, resorting to a total of 5400 aerial images and used data augmentation, in the form of a gaussian noise filter to simulate dust, although this was not considered a defect. As a result, Wang et al. achieved a classification accuracy of 94.32% and a mAP of 92.17%.

Cavieres et al. [16] projected a model capable of classifying soiling, glass breakage, cracks, defects, and faults that don't fit in the aforementioned categories were relegated to a class named other. The dataset consisted of images found through web scraping, images gathered in a PV failure laboratory,⁵ where the environmental data was also gathered, and the dataset provided by the Deep Solar Eye project [14]. Furthermore, the work used the images and the environmental data, such as the irradiance, temperature, humidity among other things to predict each module's performance individually. The proposed solution starts by segmenting the PV modules, resorting to a Mask R-CNN trained on the dataset gathered through web scouting techniques⁶ and the Deep Solar Eye's dataset. The CNN processed the segmented PV module images, while the Multi Layer Perceptron (MLP) handled the environmental data inputs. These two branches connected and used a softmax function to determine the PV module's performance. This research obtained an accuracy of 73% for power loss estimations.

³<https://aistudio.baidu.com/datasetdetail/22041>, accessed on the 17th of April 2025.

⁴<https://universe.roboflow.com/solar-abbaw/solar-msfc>, accessed on the 17th of April 2025.

⁵Located in the Universidad Técnica Federico Santa María, Santiago, Chile.

⁶images of PV modules obtained through web scraping, which were then manually labeled, using VGG annotator.

Dwivedi et al. [22] also turned to Deep Solar Eye for a dataset containing 45469 visual images of soiled and healthy solar panels, as well as images found through Google to help with other defects such as bird droppings or nests, snow coverage, cracks, shadows from trees, plants, or buildings and hardened cement, which is a total of 7 classes and 6 defects. The proposed method applies the Vision Transformer (ViT), which is an attention based deep learning model. The outputs of the transformer model are then fed to an MLP that produces a classification.

Di Tommaso et al. [23] gathered both IR and visual images of two different PV plants. The respective IR portion of the study is also mentioned in the respective thermal segment. Regarding the visual dataset, 1500 images were gathered in the PV solar plant located in Campania, Italy, with the aid of a drone. This dataset contained 6 different defects, including: puddle, soiling, strong soiling, delamination, raised panel, and bird dropping. Di Tommaso et al. employed a three-stage processing pipeline: it begins by identifying PV modules, then locates potential defects within them, and finally applies a false alarm filter to validate the detected defects. Detecting the PV modules utilizes a combination of computer vision techniques consisting of a Canny detector to detect edges, a Hough transform to detect lines, and an image rotation to horizontally align the PV modules. Detecting the defects is done through the use of a YOLOv3. The multi-stage architecture achieved a precision ranging from 73.75% (strong soiling) to 93.61% (delamination), considering an Intersection over Union (IoU) of 0.5. Note that the visual images do not aid in the detection of the defects present in the thermal images or vice-versa; in fact, the images of the datasets are taken at different solar plants.

2) THERMAL SPECTRUM

Ma et al. [25] propose an algorithm capable of detecting solar cell defects, diode failure, and occlusion using 2188 AAV-based IR images.⁷ The solution incorporates a lightweight multi-scale convolution module that combines Mobile-Net and Ghost-Net to efficiently extract features while reducing parameters and computation, replacing the original Spatial Pyramid Pooling - Fast (SPPF) module with Focal Nets and integrating a Small Object Resultful Pyramid (SORP) to enhance the detection of small defects by improving the network's ability to retain fine-grained features. The architecture presented by the authors achieved an mAP@50 of 88.7% using the YOLOv10n. Rocha et al. [26] gathered a total of 1368 thermal images from a PV plant in Beja, Portugal, with the aid of a AAV. These images were manually labeled and contained 5 defects, which are: hotspot, module in open circuit, substring in open circuit, module in short circuit, and heated module junction box. A mask region-based convolutional neural network (RCNN) was trained with a Resnet-50 as backbone using Detectron2. This solution obtained a mAP of 88% during segmentation of failures and

⁷gathered at the PV power station in Shenyang City, China.

TABLE 1. Defect detection and classification studies.

Spectrum	Author	Image Resolution	Dataset size	Method	Defects Detected	Performance
Visual	Kuo et al. [24] (2023)	227x227	600	CNN, XYZ, SGD	5	99.17% (Classification)
	Gao et al. [20] (2025)	640x640	Panel-2: 2400 Solar panel: 5443	YOLOv8-based	3	Panel-2: 87.2% Solar panel: 72 %
	Wang et al. [21] (2020)	Not available	5400	YOLOv3	3	92.17%
	Cavieres et al. [16] (2022)	224x224	260	Mask RCNN for PV segmentation, MLP and CNN for defect classification	4	73% (Classification)
	Dwivedi et al. [22] (2024)	72x72	4110	ViT	6	98.66% (Classification)
	Tommaso et al. [23] (2022)	1600x1200	1500	Computer Vision techniques and YOLOv3	6	73.75% to 93.61%
Thermal	Kuo et al. [24] (2023)	227x227	800	CNN, XYZ, SGD	7	97.52% (Classification)
	Ma et al. [25] (2025)	640x512	2188	SPPF, SORP	3	88.7%
	Rocha et al. [26] (2023)	640x512	1368	Mask RCNN	5	88%
	Zhang et al. [27] (2022)	640x512	10772	YOLOv5	6	80.88% (Classification)
	Tommaso et al. [23] (2022)	640x512	2038	Computer Vision techniques and YOLOv3	6	80.30% to 86.72%

a mAP of 96% during the segmentation of defective PV modules, both cases with an IoU of 0.5.

Zhang et al. [27] identified hotspots, sub-string abnormality, short circuit, open circuit, module position abnormality, shadow abnormality, utilizing a dataset with 10772 infrared images. The research was tested in a PV plant located in Shanxi, China, which resulted in an mAP of 80.88%, with an IoU of 0.5. Furthermore, the proposed system resorts to a YOLOv5 neural network.

Di Tommaso et al. [23] also gathered IR images in a PV plant in Sicilia, Italy. The IR portion of the dataset had 2038 thermal images, containing hotspot and junction defects. The precision for the hotspot class was 80.30%, whereas the precision for the junction class was 86.72%, both with an IoU of 0.5.

3) THERMAL AND VISUAL SPECTRUM CROSS VALIDATION

Ghahremani et al. [28] analyzed the advancements in defect detection resorting to AI solutions and did not find any studies that combined the visual and thermal spectrums for defect detection in PV panels. To the best of the author's knowledge, the research available that covered the possibility of leveraging both thermal and visual spectrums to identify and classify defects in PV panels was done by Kuo et al. [24]. Since this study considers two spectrums, there is an entry for each spectrum in the respective tables. The dataset for the RGB images contained 600 images, whereas the thermal dataset contained 800 images, which were gathered with the aid of a AAV in a Taiwan PV plant that contained 1,482 PV modules. Regarding the visual images, a total of 5 defects

were considered: dirt, vegetation, snail, hot spot, and DC box component. Regarding the thermal portion of the dataset, a total of 7 defects were considered: dirt, vegetation, thermal battery, multi-thermal battery, open circuit, cell crack, and DC box component. No data fusion was considered; the authors stitched the gathered images together through feature detection and matching to form a single aerial image of the PV plants. Sequentially, the PV modules were segmented using the Otsu algorithm [29], followed by an edge detection algorithm that relied on the Laplacian operator. Lastly, the study used a CNN classifier to find the PV module image defect features, paired up with a ReLU activation function and a cross entropy loss function. Since the dataset was also tested separately, the model achieved an accuracy of 99.17% when testing visual images and achieved 97.52% accuracy when testing thermal images, in both cases the color space XYZ and the optimizer SGD were optimal.

Analyzing the studies mentioned before, two things stand out. Firstly, the lack of available datasets, specially datasets that portray the same scene in both the IR and the visual spectrums cripples any attempt to improve the PV panel inspection procedures. Secondly, the studies mentioned don't consider the possibility of analyzing the two spectrums at the same time for defect detection. This is particularly important to note since some defects can only be detected in one of the spectrums, limiting the practicality of the solution presented in these studies. Even when both spectrums were used for cross validation, the output of the models that detected defects individually in each spectrum was compared to it's counter part to access the accuracy of the results, as opposed

to detecting defects in both spectrums in the same image, which is crucial for establishing causality relationships between defects as well as validating the information of each spectrum.

D. DEEP LEARNING MODELS

The YOLO model is a prominent object detection framework designed to perform detection in real-time by reformulating object detection as a single regression problem. Unlike traditional methods, which employ region-based or two-stage pipelines, YOLO uses a single CNN pass to predict bounding boxes and class probabilities simultaneously. This unified approach provides a substantial speed advantage, making YOLO suitable for applications requiring rapid inference, such as detecting defects in PV panels. In YOLO, the input image is divided into an $S \times S$ grid. Each grid cell predicts a predefined number of bounding boxes and assigns confidence scores to indicate the likelihood of objects in those boxes. Furthermore, class probabilities are calculated for each detection, enabling multi-class object detection. The network architecture of YOLO consists of convolutional layers designed to extract spatial features from the input, with down-sampling achieved via stride-2 convolutions rather than pooling layers. Leaky Rectified Linear Unit (ReLU) or Sigmoid Linear Unit (SiLU) (depending on the version or implementation) activation functions are used in hidden layers, while the final layer - that predicts both bounding boxes coordinates and class probabilities - use a linear activation function or a sigmoid activation function (depending on the version or implementation) for final decisions. The detection process incorporates anchor boxes, which are predefined based on statistical properties of the dataset and optimize detection across varying object sizes. Modern implementations of YOLO, such as YOLOv3, YOLOv4, and YOLOv8, have introduced various enhancements to improve accuracy and efficiency. Techniques such as multi-scale detection, cross-stage partial networks (CSPNet), spatial pyramid pooling (SPP), and path aggregation networks (PANet) have enabled YOLO to achieve significant advances in performance while maintaining its real-time capabilities [30].

The RT-DETR model introduced a cutting-edge object detection framework designed to address the computational challenges of deploying transformer-based architectures in real-time scenarios [9]. Unlike traditional detection models or earlier transformer-based systems, RT-DETR explicitly optimizes both the model architecture and processing flow to achieve low-latency object detection while maintaining competitive accuracy. RT-DETR builds upon the DETection TRansformer (DETR) paradigm, which reframes object detection as a set prediction problem, eliminating the need for non-maximum suppression (NMS) or anchor boxes that are used in the YOLO models. The model uses a hybrid efficient encoder, that takes multi-scale feature map, generated by a convolutional backbone and transforms them into a sequence of image features, afterwards these features are filtered

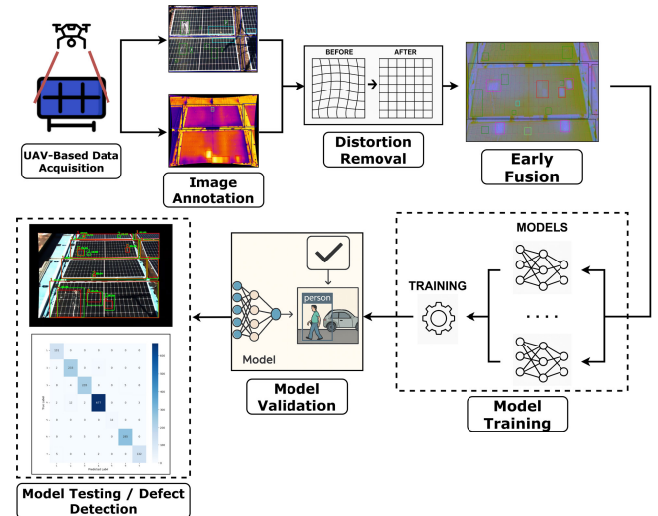


FIGURE 1. Pipeline for multispectral PV defect detection: From AAV Data acquisition to model validation and automated defect detection.

with the help of an uncertainty-minimal query selection and then the decoder (and auxiliary prediction heads) provides categories and boxes by iteratively optimizing object queries. To achieve real-time performance, RT-DETR incorporates several critical optimizations, which include lightweight backbone architectures for feature extraction, streamlined pipelines, multi-scale attention modules to balance computational efficiency with multi-scale representation capability, and parallelized decoding mechanisms to reduce latency. These advancements address the inherent computational overhead of transformers by reducing redundant computations and enhancing inference efficiency. With its ability to process visual data at high frame rates without significant loss in detection quality, RT-DETR achieves state-of-the-art balances between speed and accuracy.

III. MULTISPECTRAL DATASET STRUCTURING

Figure 1 presents the workflow of the proposed methodology. The process begins with the acquisition of the data that constitutes the dataset. Subsequently, each image is meticulously annotated consistently, adhering to the class definitions outlined in Section III-C. To ensure data accuracy, distortions introduced by the imaging sensors must be corrected in both the images and their corresponding annotations. After the dataset has been fully prepared, the models can be trained, validated and tested using a separate subset of images not involved in the training or validation phases. Ultimately, the models demonstrating the highest performance are used for defect detection in PV panels.

A. DATA ACQUISITION METHOD

The dataset consists of over 4,000 annotated images of solar panels, captured in both thermal and visual spectrums. The images were obtained at Energias de Portugal's (EDP)

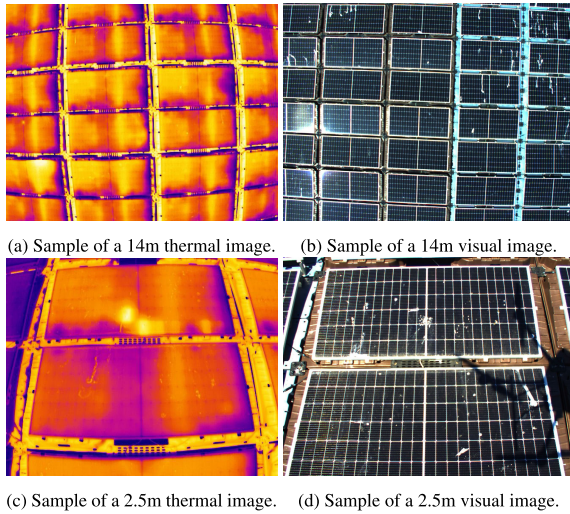


FIGURE 2. Image samples at different altitudes.

floating photovoltaic solar plant⁸ with the help of the Robotic Aerial Vehicle for any-Environment Navigation (RAVEN) [31]. A set of pictures was taken at 2.5m of altitude, with an angle of 60° in reference to the drone and the other set of images was taken at 14m of altitude with an angle of 90° in reference to the drone, which can be seen in Figure 2. The dataset was curated to aid in the development of deep learning models for the detection and classification of common issues affecting the performance and efficiency of photovoltaic (PV) systems. This PV plant has around 12 thousand PV panels, which have been placed with a 5 degree inclination in an east-to-west position to minimize wind resistance that could cause issues to the PV panels; together they produce 1.5GWh. According to the specifications described in the IEC TS 62446-3 standard [13], the IR camera must have a spectral response of 2 μm to 5 μm or 8 μm to 14 μm , a temperature-sensitivity and calibration range of -20 °C to +120 °C and an operating ambient air temperature range of -10 °C to +40 °C, among other restrictions. To encompass all these specifications, a Forward Looking Infrared (FLIR) camera was utilized, following the guidelines mentioned in the technical documentation regarding thermal imaging cameras. Furthermore, a visual camera was also attached to the drone. Additionally, GPS coordinates were also recorded to easily identify which image corresponds to which PV panel in the real world. The AAV can be seen in Figure 3, at the Alqueva floating PV solar plant.

The sensor specifications of each sensor attached to the drone are:

- **Thermal camera:** FLIR Boson 640 Radiometry camera with a frame rate of 50 Hz, resolution of 640 by 512 pixels, field of view 50° horizontal and temperature measurement accuracy of $\pm 5^\circ\text{C}$.



FIGURE 3. RAVEN flying over the Alqueva solar plant.

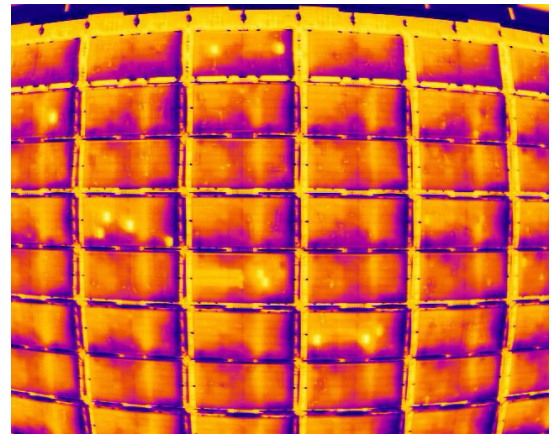


FIGURE 4. Example of image with barrel distortion.

- **Visual camera:** HIKROBOT MV-CA016-10UC camera with a frame rate of 50 Hz, resolution of 1440 by 1080 pixels and field of view of 90° horizontal.

B. IMAGE DISTORTION

The annotations were made with the help of the Computer Vision Annotation Tool (CVAT), an open-source tool developed to allow easy annotation of images through the use of bounding boxes and respective labels. The images were acquired having a negative radial displacement, i.e. Barrel Distortion, an effect much more noticeable in the thermal images. An example of this distortion is shown in Figure 4.

The calibration process was based on the implementation of Wang et al. [32], which is based on the method studied by Gramkow [33]. This process yielded the parameters necessary to remove the image distortion, including the visual vertical and horizontal focal lengths, the optical centers and the distortion coefficients of both sensors. The images were annotated with distortion to provide a more versatile dataset. Moreover, annotating the images before removing distortion also makes the dataset independent of the quality of the

⁸<https://www.edp.com/pt-pt/historias-edp/alqueva-solar-fotovoltaico-flutuante>

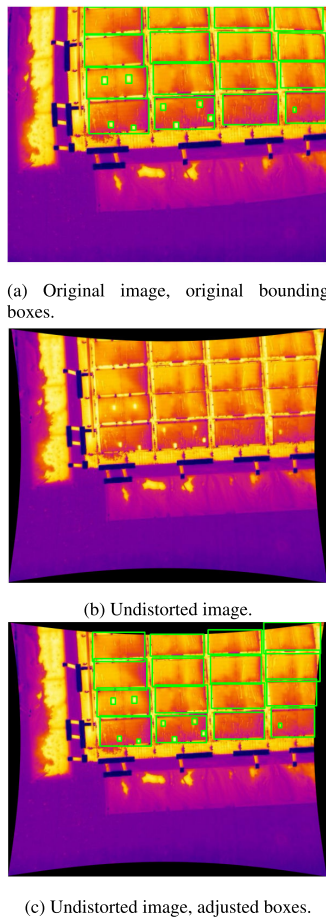


FIGURE 5. Distortion removal process.

camera calibration process. However, to produce a robust model, independent of the sensors used to gather data, the removal of distortion in both the bounding boxes and images is required.

Figure 5 shows the original image with the original annotations and the sequential process necessary to obtain the image and respective bounding boxes without distortion. Figure 5a represents the image without distortion and the original bounding boxes. Figure 5b contains the first transformation necessary, which is the removal of distortion in the image. Figure 5c shows the final result, which is the image without any distortion and the respective and adjusted bounding boxes. During the distortion removal process, some information is lost, therefore padding needs to be added to the image to maintain the original resolution and aspect ratio, which is crucial for object detection models.

C. DEFINITION OF DEFECT LABELS

This dataset is composed by three main components, which are the visual spectrum dataset, the thermal spectrum dataset, and the fusion of both datasets, the early fusion dataset. These three components may be treated as separate and unrelated datasets, which implies that each dataset has its own classes

and defects associated with it. However, some classes are the same for all three (i.e. PV module class). In total, the visual dataset has 3 classes (PV module, Soiling, Bird Drops), the thermal dataset has 4 classes (PV module, Dirty PV module, Hotspot, Heated Junction Box), and the early fusion dataset has 7 classes (PV module, Soiling, Bird Drops, Dirty PV module, Hotspot, Heated Junction Box, Hotspot Caused by Soiling/Bird Drops). The classes are as follows:

- **PV Module:** From a visual perspective, a PV module appears as a flat, rectangular panel typically composed of uniform, dark-colored PV cells arranged in a grid pattern. These cells are framed by thin metallic lines, known as busbars, that serve as electrical conduits, but their geometry adds to the module's aesthetic symmetry. The module's surface is usually covered by tempered glass to protect the cells and provide a smooth, reflective finish. They are often mounted in rows across solar plants, forming vast fields of reflective panels angled towards the sun.
- **Hotspot:** Hotspots occur when certain PV cells within a solar module experience higher temperatures than the surrounding cells, often caused by defects like short circuits, open circuits, physical damage, shading, or debris obstructing sunlight. When a hotspot forms, the affected cell no longer functions optimally and can dissipate energy as heat, which may reduce the efficiency of the entire module. Over time, this can lead to long-term damage, including cell degradation, further reducing the module's overall performance.
- **Heated Junction Box:** Heated Junction Box anomalies occur at the edges of a PV module, where the electrical connections are housed. These anomalies can result from poor soldering, faulty connections, or moisture ingress, leading to irregular electrical flow. As a consequence, temperature variations arise in the affected areas, potentially causing localized overheating. This thermal imbalance can reduce the efficiency of the module, as heat spots may interfere with normal energy flow, and in severe cases, can cause long-term damage or failure of the module's performance.
- **Residual Damage:** A Dirty PV module is a photovoltaic panel that shows temperature differences across its surface due to a lack of quick maintenance. When soiling remains in the PV module long enough to damage the cells it is affecting, but is then cleaned, which causes the defect to only be examined through a thermal image, as the soiling has been cleaned and is no longer visible to the naked eye. Over time, this temperature variation can cause thermal stress, accelerating cell degradation and potentially leading to permanent damage if not properly cleaned.
- **Bird Drops:** Bird drops are bird droppings that land on photovoltaic (PV) modules appear as white or brownish spots. These droppings can be acidic and corrosive, potentially damaging the protective coatings and photovoltaic cells of the panels over time. This

damage can lead to reduced efficiency by blocking sunlight and affecting the overall performance of the solar system. Regular inspection and prompt cleaning of bird droppings are essential to prevent long-term harm and maintain optimal energy production from the PV modules.

- **Soiling:** Soiling or dirt on PV modules occurs when dust, sand, or sediments accumulate on the surface of the solar panels. Additionally, if bird droppings are not properly cleaned, they can leave persistent residues. These defects reduce the module's efficiency by blocking sunlight and are often difficult to detect with the naked eye, especially if the buildup is subtle. Over time, soiling can lead to uneven energy production and create hotspots, which may further degrade the module's performance. Regular maintenance is essential to prevent these issues.
- **Hotspot Caused by Bird Drops/Soiling:** Soiling and bird drop patches prevent the sunlight from reaching the cells underneath them. In addition to the loss of power inherent to this, since the cells are unable to produce power, they act as a resistance to the current generated by other cells, which heats the cell, causing it damage the cell to become a hotspot and permanently damaging it [11]. These cases have been deduced from the merging process of the two datasets. If a given annotation in one spectrum is a bird drop and it coincides with a hotspot annotation, then these two annotations become one with the class Hotspot Caused by Bird Drops/Soiling.

It is important to emphasize that the entire dataset consists of images captured from a real operational PV power plant under normal working conditions, not simulated or laboratory environments. All defects identified in these images represent actual faults encountered during routine operation, making our dataset directly relevant to industrial applications. The thermal imagery annotations strictly adhere to the industry-standard IEC TS 62446-3, with defect classifications based on Annex C of that document, which provides standardized guidance for identifying faults in thermographic imagery. Visual spectrum annotations were validated by experienced site operators with expertise in PV panel maintenance, ensuring that all ground truth labels accurately represent real-world defects encountered in commercial installations.

D. DATASET METRICS

This data set has a total of 4111 images, where 2062 of these are visual and 2050 are thermal. In Table 2 both the total number of instances each class has and the number of images in which that class is present are displayed. Note that in this Table, each spectrum is being treated as a different dataset, which implies that the percentage tied to the numbers presented is encapsulated within the dataset of its respective spectrum. Furthermore, the early fusion dataset only contains images from a 2.5m altitude. An example of every class can be found in Figure 6, where the color of the bounding

TABLE 2. Characterization of the dataset.

Spectrum	Type of defect	No. of instances	No. of images	Color
Visual	PV Module	26468 (40.315%)	2062 (100%)	Dark Blue
	Dirt	4776 (7.275%)	1510 (73.230%)	Light Blue
	Bird Drops	34409 (52.410%)	2043 (99.079%)	Dark Green
Thermal	PV Module	20326 (46.897%)	1638 (79.902%)	Dark Blue
	Hotspot	9212 (21.254%)	1497 (73.024%)	Red
	Heated Junction Box	1145 (2.642%)	705 (34.390%)	Light Green
Fusion	Residual Damage	12659 (29.207%)	1954 (95.317%)	Black
	PV Module	2501 (13.495%)	1073 (90.702%)	Dark Blue
	Dirt	2463 (13.290%)	897 (75.824%)	Light Blue
	Bird Drops	7410 (39.983%)	1166 (98.563%)	Dark Green
	Hotspot	1526 (8.234%)	537 (45.393%)	Red
	Heated Junction Box	154 (0.831%)	137 (11.581%)	Light Green
	Residual Damage	3082 (16.630%)	1137 (96.112%)	Black
Fusion	Hotspot caused by Bird Drops/Dirt	1397 (7.538%)	537 (45.343%)	Magenta

box of each defect is associated to the color assigned to the respective defect in Table 2.

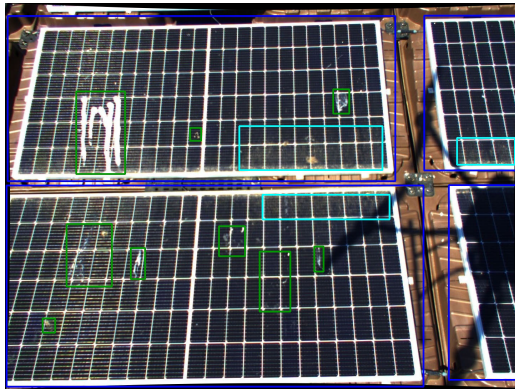
IV. MULTIMODAL DEFECT AND CLASSIFICATION ARCHITECTURE

A. FUSION STRATEGY SELECTION

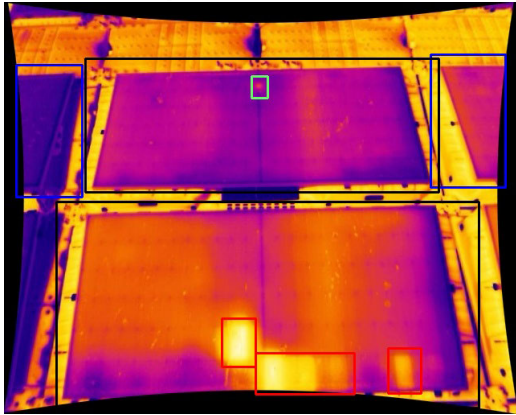
Multiple fusion strategies exist for combining multimodal data in object detection systems, generally categorized as early, mid-level, and late fusion approaches. Early fusion combines data at the input level, mid-level fusion integrates features after initial processing, and late fusion merges decisions from separate modality-specific models. This study deliberately adopts an early fusion strategy based on several key considerations specific to PV defect detection.

Early fusion in the YUV color space provides significant advantages for our application. First, it preserves the critical spatial correspondence between defects that manifest in both visual and thermal domains, such as the causal relationship between soiling (visible spectrum) and hotspots (thermal spectrum). This spatial alignment is fundamental for establishing the “Hotspot caused by Bird Drops/Soiling” defect class that represents a causal chain between visible contamination and thermal anomalies.

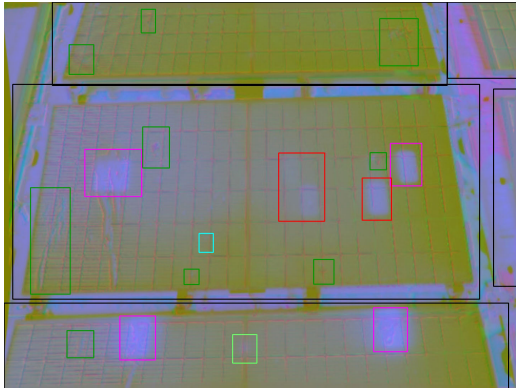
Early fusion demonstrates computational efficiency advantages over mid-level or late fusion alternatives. While mid-level fusion requires duplicate feature extraction networks running in parallel and late fusion maintains entirely separate detection pipelines, early fusion consolidates processing at the input stage. This consolidation is particularly beneficial for AAV-based inspection systems where computational resources and power are limited. As demonstrated in the performance analysis (Table 5), the early fusion approach



(a) Example of defects in the visual dataset.



(b) Example of defects in the thermal dataset.



(c) Example of defects in the early fusion dataset.

FIGURE 6. Examples of defects in the dataset.

maintains inference times comparable to single-modality processing while providing enhanced detection capabilities.

Third, early fusion facilitates end-to-end learning of cross-modal relationships by allowing the neural network to directly identify correlations between thermal signatures and visual features during training, rather than relying on engineered feature combination strategies that might not optimally capture these relationships.

While mid-level fusion could potentially offer more specialized feature extraction for each modality before combination, our experiments indicated that the YUV-based

early fusion approach provided sufficient modality-specific information preservation while maintaining model simplicity. Late fusion would eliminate the need for precise spatial alignment between modalities but would sacrifice the ability to learn joint representations of defects that manifest across both spectra.

Given these considerations and our focus on developing a system suitable for real-time AAV-based inspection, early fusion presented the optimal balance between computational efficiency, detection accuracy, and the ability to establish cross-modal defect relationships.

B. PRE-PROCESSING IMAGES FOR EARLY FUSION

The fusion of the IR and visual datasets was accomplished by spatially and temporally synchronizing each image. During the acquisition process, each image acquired by each sensor was labeled with the respective timestamp of its acquisition. Using these timestamps, images were synchronized with respect to time. Spatially synchronizing the two images requires finding the homography between the two images. The homography describes the relationship between the two perspective image planes and defines a projective transformation matrix that, when applied to a given image taken from a sensor, transforms its perspective into the target sensor's perspective. Furthermore, from a coordinate point of view, Equation 1 governs this concept,

$$s \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (1)$$

where the matrix H transforms a 2D point $[x', y']^T$ into point $[x, y]^T$ [34]. In this case, since the sensors are mounted to the drone and have the same angle in relation to the drone, the only difference between their perception of the real world is imposed by the physical distance that separates the two sensors, which translates to the homography through a translation of one image into the other.

To effectively detect defects across both spectral domains, the visual and thermal images must be spatially aligned with sub-pixel precision. As the annotations are performed before the fusion process, the early fusion modality inherently results in two separate annotations per object—one from each spectrum—representing the same physical entity. To optimize training and evaluation of the AI models, these duplicate annotations must be merged during the fusion stage, ensuring that each object is represented by a single, unified annotation. For instance, a solar panel annotated separately in both the visual and thermal spectra will be combined into a single annotation in the fused image. This precise alignment is particularly critical for establishing causality relationships between defects, such as the “Hotspot Caused by Bird Drops/Soiling” class that represents a direct connection between visual contamination and thermal anomalies. In our application, alignment errors exceeding 3 pixels resulted in over 40% reduced detection accuracy for causally-related defects during preliminary testing. The barrel distortion

present in thermal images (shown in Figure 4) further complicates this alignment process, requiring the distortion removal procedure detailed in Section III-B before applying the homography transformation. Without this level of precision, defect detection becomes unreliable or impossible, as the model cannot correctly associate thermal signatures with their corresponding visual features, making accurate fusion-based detection infeasible.

Once the homography was obtained, our fusion approach leverages the YUV color space to effectively combine the complementary information from both sensing modalities. The YUV color space was specifically selected because it separates luminance (Y channel) from chrominance (U and V channels), as well as the UV channels being photometric invariant [35], allowing for a homogeneous integration of thermal and visual data compared to the RGB color space. The fusion process follows these steps:

- 1) The visual RGB image is first converted to the YUV color space using standard conversion matrices.
- 2) Since the thermal infrared image inherently captures intensity variations that correspond to temperature differences, we use this thermal image to provide the luminance information (Y channel) in our fused representation.
- 3) The chrominance components (U and V channels) from the converted visual image are preserved to maintain the color and texture information critical for identifying visual defects such as soiling and bird droppings.
- 4) The final fused image is constructed by combining the thermal image data (as the Y channel) with the U and V channels from the visual image.

This approach ensures that the fused image retains both the luminance of the IR image, providing the temperature gradient information necessary to identify defects in the IR spectrum, and the texture of the visual image, preserving the features inherent to visual defects. The YUV fusion method enables our detection models to simultaneously learn from both thermal and visual features within a unified representation, facilitating the detection of defects that manifest in either or both modalities.

Since the image needed to retain both the luminance of the IR image and the texture of the visual image, the YUV color space was adopted. The brightness channel (Y) is relegated to the IR image, providing the luminance necessary to identify defects in the IR spectrum, whereas the two chroma channels (U and V) are saved for the Green and Blue channels of the visual image, providing texture inherent to visual features.

C. MODEL SELECTION: BALANCING SPEED AND ACCURACY

When selecting appropriate deep learning models for PV defect detection in aerial inspections, striking a balance between detection accuracy and computational efficiency is crucial. Our selection process prioritized models capable of operating in real-time scenarios while maintaining high detection quality for various types of defects.

After evaluating multiple state-of-the-art object detection architectures, RT-DETR and YOLO emerged as the most promising candidates. While YOLO models are generally known for their faster inference times, as demonstrated in our experiments presented in Table 5, RT-DETR provided superior detection quality with only a marginal increase in computational overhead [36]. Specifically, on a GeForce RTX 2060 GPU, RT-DETR-X showed inference times of 61.12 ms for fusion data compared to YOLO11x's 49.12 ms, yet delivered higher mAP (0.80456 vs. 0.76165), IoU (0.91037 vs. 0.90028), GIoU (0.91249 vs. 0.89938), and mAP@75 (0.85963 vs 0.81922) metrics as shown in Table 4.

This trade-off was deemed acceptable for our application, as the approximately 12ms increase in processing time did not significantly impact the real-time capabilities required for AAV-based inspection of PV installations, while providing substantial improvements in detection accuracy. Notably, RT-DETR's transformer-based architecture demonstrated superior feature extraction capabilities, particularly important for distinguishing between visually similar defects such as hotspots and heated junction boxes.

Our selection criteria prioritized detection accuracy for critical PV defects that could lead to significant power loss or safety hazards, while ensuring the system remained practical for field deployment. The decision to include both models in our evaluation framework also provides flexibility for practitioners to select the most appropriate model based on their specific hardware constraints and accuracy requirements.

D. PRACTICAL IMPLEMENTATION AND RISK MITIGATION

The transition from theoretical model development to practical field implementation presents several technical challenges that our approach systematically addresses. First, sensor alignment and calibration between thermal and visual modalities are critical for accurate fusion. Our homography-based calibration process, detailed in Section III-B, accounts for lens distortion and sensor positioning differences, ensuring reliable data fusion even with commercially available drone-mounted sensors that may have different fields of view and resolutions.

Processing power constraints are particularly relevant for AAV-based inspection systems due to weight, power, and size limitations. As demonstrated in Table 5, our comparative analysis of model performance on consumer-grade hardware (RTX 2060) shows that real-time processing is achievable with both models, with RT-DETR-X operating at 16-21 FPS and YOLO11x at 20-29 FPS, making practical field deployment feasible with relatively modest computational resources.

The problem of false positive detections, which can lead to unnecessary maintenance interventions and increased operational costs, is mitigated through our fusion approach. By requiring consistent detection evidence across both spectral domains for certain defect types (particularly for the

“Hotspot caused by Bird Drops/Soiling” class), the system inherently reduces false positives that might occur when analyzing either spectrum independently.

Finally, our system provides defect classifications aligned with industry standards (IEC TS 62446-3), facilitating direct integration with existing maintenance protocols and documentation requirements. This standardization reduces implementation barriers and ensures that detection results can be immediately actionable within established maintenance workflows.

V. RESULTS

From the dataset described in III, only the images collected with an altitude of 2.5m, for a total of 1183 images, were used during training and testing. Initially the dataset was divided in three components, 80% (947 images) for training purposes, 10% (118 images) for validation of the training and the remaining 10% (118 images) for testing purposes, while ensuring the class proportionality in each component, meaning that each class instance was equally distributed among all three components. The training dataset was augmented by resorting to computer vision techniques, including rotations, bidirectional flips, ISO noise, Median Blur, and Embossing. To account for the varying angles and orientations from which AAVs capture images of PV panels, the datasets underwent augmentation through image transformations. While vertical and horizontal flipping was implemented, color-altering augmentations were deliberately avoided to preserve defect textures, particularly for hotspot defects where intensity characteristics are crucial. Instead, the augmentation process incorporated ISO noise and Gaussian Blur, as these transformations maintain the essential features of PV panel defects. Several AI models were trained and tested using the fusion modality, namely the FasterRCNN, Fully Convolutional One-Stage Object Detection (FCOS), RetinaNet, Single Shot multiBox Detector (SSD) and SSDLite, RT-DETR-X, and YOLO11x. For this purpose, the same models were trained and tested using the dataset encompassing only one spectrum, either the visual or the IR dataset, using the same methodology and data augmentation techniques. The models maintained the default parameters during training, including the loss functions, meaning that no fine-tuning was employed; images and labels were fed forward following each model’s architecture and parameters. During training, the top performing models were saved based on the validation loss. The common hyperparameters used across all models are detailed in Table 3, which specifies key parameters such as learning rate, batch size, optimization algorithm, training duration, and thresholds for both confidence and IoU. The learning rate of 0.001 balances convergence speed and training stability. The IoU threshold of 0.5, which is a standard value that ensures accurate object localization while accommodating minor variations in boundary predictions, defines the minimum required overlap between predicted and actual bounding boxes for valid detections. A confidence threshold of 0.4 was chosen to filter

TABLE 3. Training Hyper-parameters.

Parameter	Value
Epochs	300
Seed	7
Early Stopper Patience	50
Batch Size	2
Image Size	640
Learning Rate	0.001
Confidence Threshold	0.4
IoU Threshold	0.5
Optimizer	AdamW

out low-confidence predictions while maintaining detection sensitivity. The optimizer used was AdamW [37], which is a stochastic optimization method typically used for object detection algorithm. The experiments were conducted on two different GPU platforms to evaluate both training and inference performance. A consumer-grade Nvidia GeForce RTX 2060 with 6GB of VRAM was used primarily for testing and inference operations, providing a more accessible hardware configuration that could be used in real-world applications. For additional inference benchmarking, an Nvidia A100 with 40GB of VRAM was used, enabling comparative performance analysis across different hardware configurations.

Early fusion techniques in object detection enable the integration of multimodal data during the initial processing phases, potentially enhancing detection accuracy and robustness. The effectiveness of this methodology was assessed through a thorough comparative analysis of leading object detection models, using key performance indicators including mean Average Precision (mAP), Intersection over Union (IoU), and Generalized IoU (GIoU).

Before analyzing the performance results, the architectures of the two main models used in this study must be showcased. Figure 7 shows the RT-DETR, which consists of a Backbone for initial feature extraction, followed by an Efficient Hybrid Encoder that processes multi-scale feature maps. This encoder integrates Aligned Feature Fusion Injection (AIFI) modules and Complementary Channel-wise Feature Fusion (CCFF) to effectively combine features from different scales. The uncertainty-minimal query selection module then filters the most reliable feature candidates, which are processed by the decoder and detection head to generate the final bounding box predictions and classifications. Figure 8 illustrates the YOLOv11 architecture, featuring its three core components: Backbone, Neck, and Head. The Backbone handles multi-scale feature extraction using advanced blocks such as the C3K2 and Spatial Pyramid Pooling Fast (SPPF), designed for efficient feature representation. The Neck aggregates and refines these features with additional mechanisms like Cross-Stage Partial with Spatial Attention (C2PSA). Finally, the Head predicts object bounding boxes, masks,

TABLE 4. Methods comparison for early fusion.

Model	Modality	mAP	IoU	GIoU	mAP50	mAP75
FasterRCNN	V+T	0.40787	0.77514	0.77580	0.68385	0.38360
FCOS	V+T	0.17151	0.71037	0.72120	0.26821	0.17804
RetinaNet	V+T	0.18908	0.70545	0.72880	0.26446	0.21407
SSD	V+T	0.09468	0.65693	0.66242	0.27543	0.04392
SSDLite	V+T	0.27029	0.72060	0.73529	0.40580	0.27805
RT-DETR-X	T	0.56168	0.89055	0.88934	0.80128	0.57450
	V	0.74517	0.89163	0.89427	0.92596	0.77643
	V+T	0.80456	0.91037	0.91249	0.94198	0.85963
YOLO11x	T	0.59618	0.89056	0.88982	0.83846	0.63357
	V	0.76333	0.89752	0.89717	0.93391	0.79118
	V+T	0.76165	0.90028	0.89938	0.94976	0.81922

and classifications using enhanced multi-scale processing. These architectural differences are key to understanding the performance variations discussed in the following analysis.

Figure 9 shows the training and validation loss curves for the RT-DETR-X and YOLO11x models when applying early fusion, tracked over 225 epochs. Both models exhibit initially high losses that rapidly decrease in the first 50 epochs. After epoch 50, the RT-DETR-X model begins to stabilize while the YOLO11x model only stabilizes at epoch 75. Interestingly, YOLO11x demonstrates a smaller gap between its training and validation losses, while RT-DETR-X shows a more pronounced difference between its training and validation performance. Looking at the training curves, there are no signs of overfitting in either Fusion model, as evidenced by the continuing decrease in training loss while the validation loss remains relatively stable after epoch 50. This consistent divergence between training and validation performance during the entire training process suggests the models are memorizing learning generalizable features rather than the training data.

The performance results conducted on the test dataset for both the RT-DETR-X and the YOLO11x models are presented in Table 4, whereas the inference statistics are presented in Table 5. In these and following tables, the modality of the results obtained are referred to as V, T or V+T, where V refers to only the visual dataset, T refers to only the thermal dataset, and V+T refers to the early fusion dataset. Likewise, the inference time and statistics of each model is revealed in Table 5, also follow this structure.

Based on the experimental results presented in Table 4, one can observe that RT-DETR-X and YOLO11x significantly outperform other architectures in the early fusion scenario. RT-DETR-X demonstrates high performance across all metrics, achieving the highest scores in IoU (0.91037) and GIoU (0.91249), mAP (0.80456), and mAP75 (0.85963), indicating superior accuracy in object localization and boundary prediction. Similarly, YOLO11x shows remarkable consistency with high scores across multiple metrics, particularly in mAP50 (0.94976). Given their performance

compared to traditional architectures like FasterRCNN, FCOS, and RetinaNET, the focus of the subsequent analysis, particularly the impact of the early fusion method, lies on RT-DETR-X and YOLO11x as they represent the most promising approaches for early fusion-based object detection in our application context.

Firstly, regarding RT-DETR-X, three distinct scenarios were evaluated: single-modality Visual input, single-modality Thermal input, and a multimodal approach combining both Visual and Thermal inputs through fusion. Furthermore, the inference time for the RT-DETR-X neural network, when detecting all classes, is also shown in Table 5. Looking at the results present in Table 4, several interesting patterns emerge. The Fusion approach demonstrates superior performance compared to single-modality methods, which means that it is leveraging complementary information from both Thermal and Visual inputs. This complementarity is particularly evident in the IoU and GIoU metrics, where Fusion models achieve scores of 0.91037 and 0.91249, respectively, indicating good object localization accuracy. Notably, the Visual model performs well on its own, especially in mAP50, where it outperforms Fusion with a score of 0.92596. This suggests that while Fusion generally provides more robust detection, the Visual modality remains highly effective for less strict detection thresholds. The Thermal spectrum, while showing decent performance, consistently scores lower across all metrics, particularly in mAP and mAP75 (0.56168 and 0.57450, respectively). This performance gap indicates that thermal imaging alone may struggle with precise object detection and classification, especially at higher IoU thresholds. However, its integration into the Fusion system contributes positively to overall performance, suggesting it provides valuable complementary information that enhances detection robustness.

The confusion matrix shown in Figure 10 illustrates the classification performance of the RT-DETR-X model across different defect types in solar panel inspection. In this matrix, each number corresponds to: Hotspot, Dirt, PV Module, Bird Drops, Heated Junction Box, Residual Damage, Hotspot caused by Bird Drops/Dirt. The model demonstrates particularly strong performance in detecting Bird Drops (class 4), achieving 677 correct classifications (97.27%), the highest among all categories. Solid performance is also seen in detecting Residual Damage (class 6: 285 instances, 98.28%), PV Modules (class 3: 235 instances, 96.31%), and Dirt (class 2: 233 instances, 95.49%). Several interesting misclassification patterns emerge from the analysis. The model shows minor confusion between PV Module (class 3) and Residual Damage (class 6) with 5 and 5 misclassifications, respectively, and between Bird Drops (class 4) and Dirt (class 2) with 12 and 9 misclassifications, respectively. The Heated Junction Box (class 5) category, despite having the fewest samples has 14 correct classifications with perfect precision (100%) and no false positives. In the case of Hotspots (class 1, 100%) and Hotspots caused by Bird Drops/Dirt (class 7, 94.29%), some confusion exists with

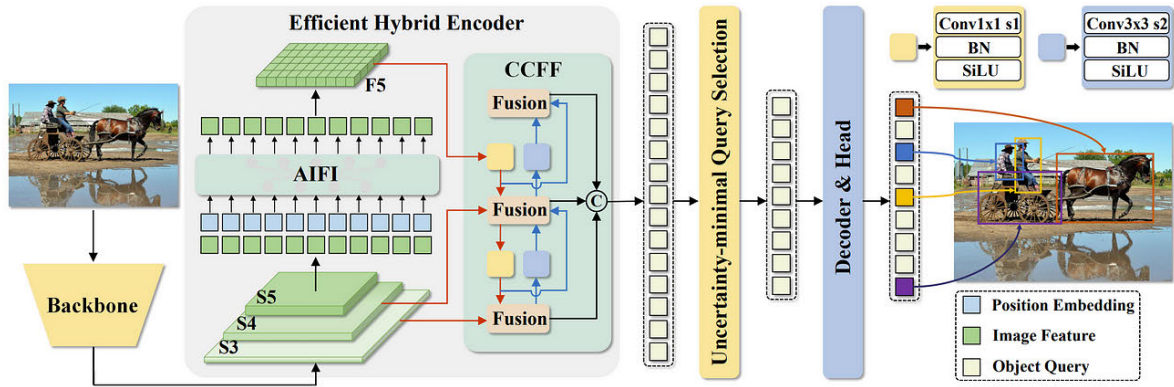


FIGURE 7. RT-DETR model architecture [9].

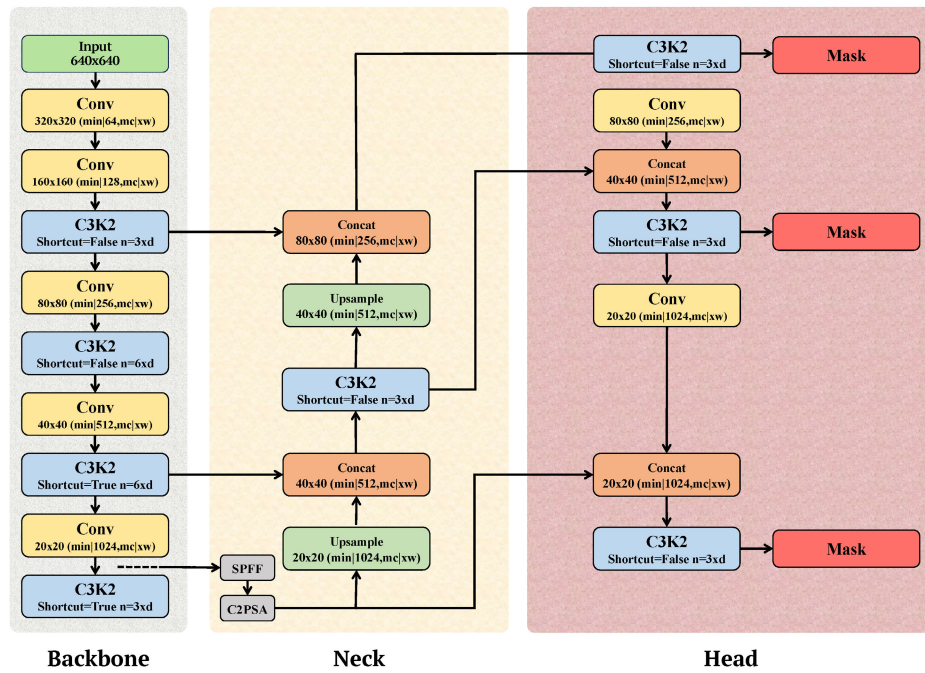


FIGURE 8. YOLOv11 model architecture [38].

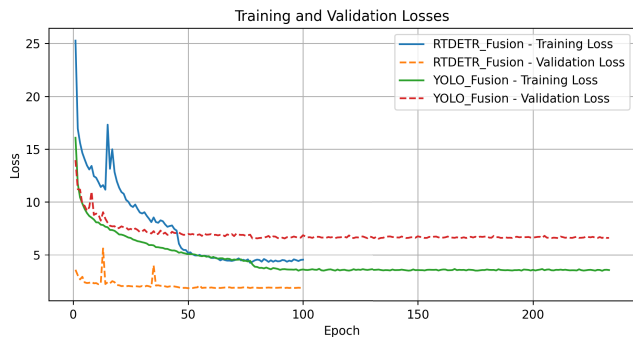


FIGURE 9. Training and Validation losses for RT-DETR-X and YOLO11x.

5 instances of the latter being misclassified as simple Hotspots (class 1), which is understandable given their

shared thermal characteristics. The strongly diagonal nature of the confusion matrix indicates robust overall classification performance, although there remains room for improvement in distinguishing between defect types that share similar visual or thermal signatures.

Regarding YOLO11x, Table 4 shows the results obtained. Similarly to RT-DETR-X, three different cases were studied, where each focuses on the different modalities applied. Furthermore, the inference time for the YOLO11x, when detecting all classes, is also shown in Table 5. Lastly, the confusion matrix is also shown in Figure 11.

The inference performance of RT-DETR-X was evaluated on two different GPU platforms: the consumer-grade GeForce RTX 2060 6GB and the data center-focused NVIDIA A100 40GB, as shown in Table 5. The results

TABLE 5. Inference statistics.

Model	GPU	Modality	Mean	Std. Dev.	Min.	Max.	Median	FPS
RT-DETR-X	Geforce RTX 2060 6GB	V	61.10 ms	1.33 ms	60.36 ms	100.61 ms	60.97 ms	16.37
		T	45.81 ms	3.24 ms	44.44 ms	138.52 ms	45.30 ms	21.83
		V+T	61.12 ms	0.63 ms	60.43 ms	71.85 ms	61.02 ms	16.36
	NVIDIA A100 40GB	V	30.87 ms	1.02 ms	30.00 ms	38.05 ms	30.52 ms	32.39
		T	31.86 ms	3.42 ms	28.62 ms	61.19 ms	31.51 ms	31.39
		V+T	30.56 ms	0.86 ms	29.75 ms	37.19 ms	30.32 ms	32.72
YOLO11x	Geforce RTX 2060 6GB	V	49.00 ms	0.25 ms	48.40 ms	52.76 ms	48.97 ms	20.41
		T	33.90 ms	6.49 ms	30.99 ms	142.47 ms	32.8 ms	29.50
		V+T	49.12 ms	0.23 ms	48.63 ms	53.14 ms	49.11 ms	20.36
	NVIDIA A100 40GB	V	15.30 ms	0.37 ms	15.08 ms	21.82 ms	15.20 ms	65.34
		T	14.57 ms	0.49 ms	13.97 ms	20.75 ms	14.50 ms	68.65
		V+T	16.21 ms	1.20 ms	15.46 ms	26.64 ms	15.82 ms	61.71

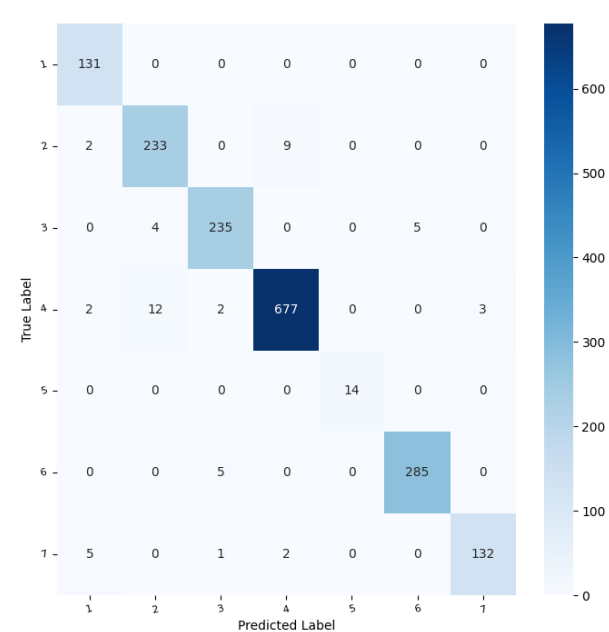


FIGURE 10. RT-DETR-X Confusion matrix.

show a significant performance difference between the two hardware configurations. The A100 demonstrates superior performance across all data types, achieving approximately twice the speed of the RTX 2060, with frame rates around 32 FPS compared to the RTX 2060’s 16-21 FPS. For the RTX 2060, thermal data processing is notably faster (45.81 ms mean) compared to visual and fusion processing (61.10 ms and 61.12 ms, respectively). Interestingly, the A100 shows more consistent performance across all data types, with mean inference times ranging from 30.56 ms to 31.86 ms. The standard deviation values also indicate that the RTX 2060 provides more stable performance, particularly for fusion data (0.63 ms) compared to the A100 (0.86 ms). This suggests that while both GPUs can handle real-time processing, the A100 offers higher performance for

production deployment, at a much higher deployment cost and slightly less reliability.

When analyzing the inference performance of YOLO11x across different GPUs, there are notable differences in processing power, demonstrated in Table 5. Comparing the GeForce RTX 2060 6GB to the NVIDIA A100 40GB, the A100 shows significantly superior performance, achieving more than three times the frame rates of the RTX 2060 across all data types. On the RTX 2060, thermal processing shows the best performance with an average of 33.90ms (29.50fps), while visual and fusion processing are slower at approximately 49ms (approximately 20fps). The A100 shows exceptional performance with consistently low inference times across all data types, ranging from 14.57 ms to 16.21 ms, which translates to frame rates between 61-68 FPS. Interestingly, YOLO11x shows lower standard deviation values on the RTX 2060 for visual and fusion data (0.25 ms and 0.23 ms, respectively) compared to thermal data (6.49 ms), demonstrating more stable performance for non-thermal processing. The A100’s performance remains highly consistent across all modalities, with slightly higher variance in fusion processing (1.20 ms standard deviation), but still excellent overall performance. This suggests that YOLO11x is well optimized for high-end GPU architectures, making it particularly suitable for real-time applications when used on powerful hardware such as the A100, but still performs quite well on less powerful hardware such as the RTX 2060.

Analyzing the YOLO11x results in Table 4 demonstrates distinct behavioral patterns and effectiveness levels across each modality configuration. The Visual and Fusion models show remarkably similar performance patterns, with Fusion achieving better scores in IoU (0.90828) and GIoU (0.89938) compared to Visual’s 0.89752 and 0.89717, respectively. Interestingly, the Visual model performs slightly better than the Fusion model in mAP, with 0.76333 versus 0.76165, suggesting that multimodality does not always guarantee improved performance in all metrics. The Thermal model shows a distinct pattern - while it has lower performance

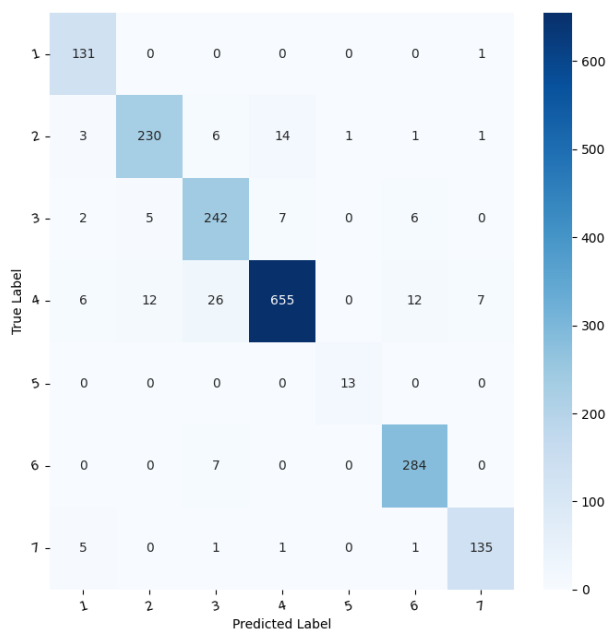


FIGURE 11. YOLO11x confusion matrix.

in most metrics, particularly in mAP75 (0.63357), it still achieves similar IoU and GIoU scores (0.89056 and 0.88982). This indicates that thermal detection can maintain good localization accuracy even when overall detection confidence is lower. The high mAP50 scores across all spectrums (ranging from 0.83846 to 0.94976) suggest that YOLO11x is particularly effective at object detection when using a more lenient overlap threshold, with the Fusion model performing best in this metric.

The confusion matrix shown in Figure 11 demonstrates the classification performance of YOLO11x. The model shows robust performance across several defect categories, with Bird Drops (class 4) achieving the highest number of correct classifications at 655 instances (91.23%). The model also performs well in detecting Residual Damage (class 6: 284 instances, 97.59%), PV modules (class 3: 242 instances, 92.37%), and Dirt (class 2: 230 instances, 89.84%). Several notable misclassification patterns emerge from the analysis. For example, there are 14 instances where Dirt (class 2) is misclassified as Bird Drops (class 4), and 26 cases where Bird Drops (class 4) are confused with PV Module (class 3). The model shows good performance with Hotspots (class 1), correctly identifying 131 instances (99.24%) but experiencing some confusion with Hotspots caused by Bird Drops/Dirt (class 7). While the Heated Junction Box (class 5) category has relatively few samples (13 correct classifications), it shows perfect precision (100%) with no false positives. The Hotspot caused by Bird Drops/Dirt (class 7) category achieves 135 correct classifications (94.41%), though it experiences minor confusion with regular Hotspots (class 1) with 5 misclassifications. The strongly diagonal nature of the matrix indicates good overall classification performance, though there remains room for improvement in distinguishing between visually similar defect types.

Overall, the results obtained are very promising, as both models have surpassed the 90% threshold in the mAP@50, with the YOLO11x outperforming the RT-DETR-X. However, the RT-DETR-X presents a higher mAP, IoU, GIoU, and mAP@75 which indicates that this model is more confident in its predictions. These metrics are important to understand as they help the PV panel maintainer to decide which model is more suitable for their particular PV plant - a model that is capable of detecting more defects but is less confident and makes more mistakes, or a model that is much more conservative in its guesses but also more confident and more often correct.

The inference performance analysis of both RT-DETR-X and YOLO11x models across different GPU platforms reveals significant insights into their operational capabilities. When comparing their performance on the consumer-grade GeForce RTX 2060 6GB and the data center-focused NVIDIA A100 40GB, YOLO11x shows superior speed capabilities across both platforms. On the RTX 2060, while RT-DETR-X achieves 16-21 FPS with thermal processing being the fastest at 45.81 ms mean inference time, YOLO11x achieves better performance with 20-29 FPS, with thermal processing reaching 33.90 ms mean inference time. This performance difference becomes even more pronounced on the A100, where RT-DETR-X operates at around 32 FPS across all data types (30.56-31.86 ms), while YOLO11x achieves more than double that performance with 61-68 FPS (14.57-16.21 ms). In terms of stability, both models show interesting characteristics: RT-DETR-X demonstrates more consistent performance across modalities on the A100, particularly in fusion processing (0.86 ms standard deviation), while YOLO11x shows varying stability patterns with better consistency in visual and fusion processing on the RTX 2060 (0.25 ms and 0.23 ms standard deviations, respectively) but higher variance in thermal processing (6.49 ms). These results indicate that while both models are capable of real-time processing, YOLO11x offers superior speed performance, particularly on high-end hardware, making it especially suitable for applications where processing speed is critical. However, RT-DETR-X's more consistent performance across modalities might be advantageous in scenarios where stability is prioritized over raw speed.

The similarities between defects become clear when analyzing the confusion matrices. Bird droppings and dirt are very similar, with similar aspect ratios and locations in each image, which leads the models to often mistake one for the other. Furthermore, PV modules and Residual damage have the most in common, considering that they are both delimiting a photovoltaic module, with the key difference being that one is a healthy PV module and the other is a damaged PV module that contains residual bird droppings or soiling that permanently damaged the underlying cells. Hotspots and hotspots caused by bird droppings or soiling can be mistaken as they have the same aspect ratios and locations, the only difference between them being whether or

not there are bird droppings or soiling that caused the hotspot. However, due to its unique and constant location, aspect ratio, and size, the heated junction box has not been a seed of doubt in the model's reasoning, which is surprising considering the similar texture to a regular hotspot.

Environmental variability presents significant challenges for deployment across diverse PV installations globally. To address these challenges, continual learning approaches show particular promise. Recent advances, such as Selective Multi-Task Coordination for Continuous Learning (SMTC-CL) [39] demonstrate how models can adapt to new conditions while preserving performance on previously learned tasks.

VI. CONCLUSION

This study begins by creating a standardized and robust dataset based in real-world images of a PV power plant. The heterogeneous information provided by the IR and RGB sensors can be combined into a single 3-channel image to allow multispectral defect detection in PV panels, resorting to the YUV color space as proposed by this article. This approach advantages defect detection in PV panels by raising the complexity of the model to enable multimodal defect detection and the establishment of causality relationships between defects in different modalities. Furthermore, results show that this approach also benefits the models, as it improves their performance in defect detection tasks when using the fused version of the dataset.

The conducted experiments reveal that AI models are capable of detecting defects in a multimodal manner with an mAP of almost 95% in the case of the YOLO11x model and 94% in the case of the RT-DETR model, given an IoU threshold of 0.5. Despite the mAP@50 being the only metric where the YOLO11x beats the RT-DETR, in other metrics, including mAP, IoU, GIoU, and mAP75, the RT-DETR outperforms the YOLO11x model. Furthermore, using the early fusion method improved defect detection by 2% comparing to the visual modality and 14% (or 12% considering the YOLO11x model) comparing to the thermal modality. Regarding inference times, RT-DETR-X was slower in all the scenarios tested by a margin of 12ms across all three modalities when using the NVIDIA GeForce RTX 2060 6GB, making YOLO11x the faster model. The thermal modality is the quickest to infer, since the images have a smaller resolution and only one channel compared to the higher resolution 3-channel visual images.

While this study demonstrates the effectiveness of multispectral fusion for PV defect detection, several promising research directions warrant further investigation. First, expanding the dataset to include additional defect types and environmental conditions would enhance model robustness across diverse operational scenarios. Second, exploring more sophisticated fusion architectures, such as attention-based mechanisms that dynamically weight the importance of thermal versus visual features depending on the defect type, could further improve detection accuracy.

Similarly, domain-specific large models like Virtual Signal Large Model (VSLM) [40] could address the class imbalance in our dataset, particularly for specialized defects like heated junction boxes. VSLM's capability to generate synthetic but physically plausible defect signatures could enhance the detection of narrow defects and augment our limited samples of rare defect types without extensive retraining.

Additionally, developing techniques for temporal analysis across multiple inspection missions could enable predictive maintenance by tracking the progression of defects over time, while implementing embedded edge computing solutions for real-time processing onboard AAVs remains a significant technical challenge requiring optimization of model architectures for low-power, weight-constrained platforms.

ACKNOWLEDGMENT

(Lourenço Sousa Pinho and Tiago Daniel Sousa contributed equally to this work.)

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